



DX-NR02

2.4G module technical

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Updated records

Version	Date	Instructions	Author
V1.0	2024/01/01	Initial version	YXR
V1.1	2024/02/10	New hardware section chart	YXR

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1. Module Introduction

1.1. Overview

DX-NR02 is based on Si24R1 as the core research and development of SHEN ZHEN DX-SMART TECHNOLOGY CO.,LTD, with high speed, high stability of small size 2.4GHz wireless transceiver radio frequency module. The module comes with high performance PCB antenna, using industrial grade 16MHz crystal, working in the ISM frequency band of 2.4GHz~2.525GHz, suitable for a variety of Internet of things occasions: smart home, wireless remote control somatosensory equipment, wireless keyboard and mouse, etc., is an ideal product for Internet of things applications.

NR02 is a pure radio frequency transceiver module, which needs to be driven by MCU or use a special SPI debugging tool.

1.2. Features

- The operating frequency band is 2.4~2.525GHz, with a total of 126 channels, which meets the application requirements of multipoint communication, grouping, frequency hopping and so on
- It supports global license-free ISM 2.4GHz band operation
- Support air rate :2Mbps/1Mbps/250Kbps
- Industrial crystal: 16MHz 10ppm
- Volume size: 15×25MM
- Under ideal conditions, the communication open and visible distance can reach 300M
- It supports 2.0-3.6V power supply
- The maximum transmit power can reach 7dbm
- Ultra-low sleep power consumption
- SPI communication: Connect with MCU through SPI interface, rate 0 ~ 10Mbps
- Equipped with PCB board antenna, reserved external antenna interface



1.3. Applications

- Wearable devices;
- Smart home and industrial sensors;
- Security systems, positioning systems;
- Wireless remote control, UAV;
- Wireless game remote control;
- Healthcare products;
- Wireless voice, wireless headphones;
- Child finder and tracker;
- Logistics tracking, warehouse inspection, electronic tags;
- Automotive industry applications.

1.4. Basic parameters

Table 1: Table of limit parameters

Main parameters	Performance		Remarks
	Minimum	Maximum	
Supply voltage (V)	0	3.6	Anything over 3.6V risks burning the module
Blocking power (dBm)	-	10	-
Operating temperature (° C)	-40	85	Industrial grade



Table 2: Working Parameter Sheet

Main parameters	Performance			Notes
	Minimum	Typical value	Maximum value	
Operating voltage (V)	2.0	3.3	3.6	Exceeding 3.6V has the risk of burning the module
Operating frequency band (GHz)	2.4	-	2.525	ISM band
Receiver sensitivity (dBm)	-96	-87	-83	250KHz, 1MHz, 2MHz
Air rate (bps)	250k	-	2M	-
Maximum transmit power (dBm)	-12	-	+7	-
Transmit current (mA)	-	25	-	7dBm&250K
Received current (mA)	-	14	-	250K
Dormant current (μA)	-	1	-	-



2. Application interface

2.1. Module pin definition

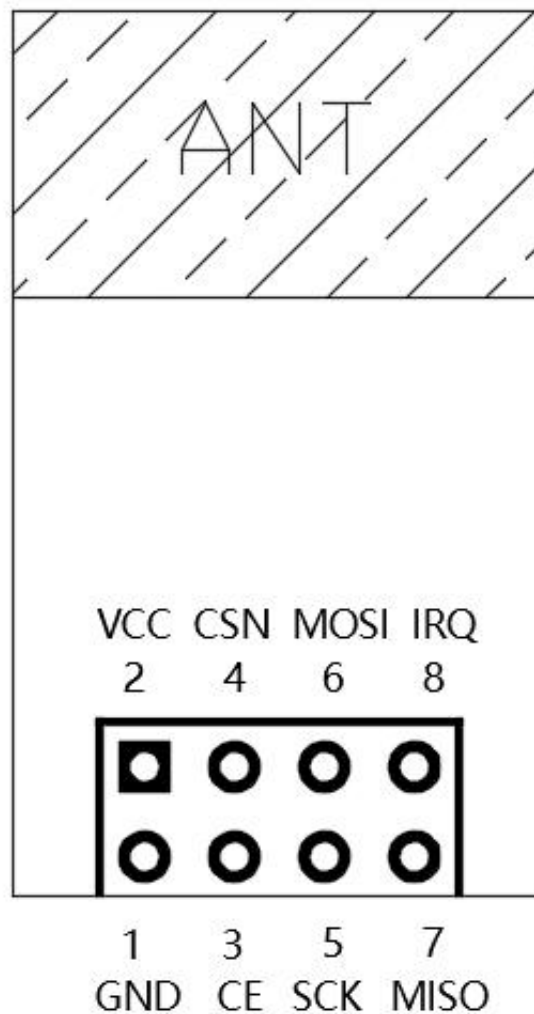


Figure 1: Module pin definition



2.2. Illustration of the pin definition

Table 3: Pin Definition Instructions Table

Pin serial number	Pin name	Pin function	Instructions
1	GND	GND	Ground
2	VCC	Power input pin	3.3V (typical value)
3	CE	Module enables control pins	-
4	CSN	Select the pin of the module piece	-
5	SCK	Module SPI bus clock	-
6	MOSI	Module SPI data input pin	-
7	MISO	Module SPI data output pin	-
8	IRQ	Module interrupt signal pin, low valid	-

2.3. Power interface design

Table 4: Power interface pin definition Table

Pin names	Pin number	Description	Minimum	Typical value	Maximum value	Units
GND	1	to	-	0	-	V
VCC	2	Module power supply	2.0	3.3	3.6	V



2.4. Reference connection circuit

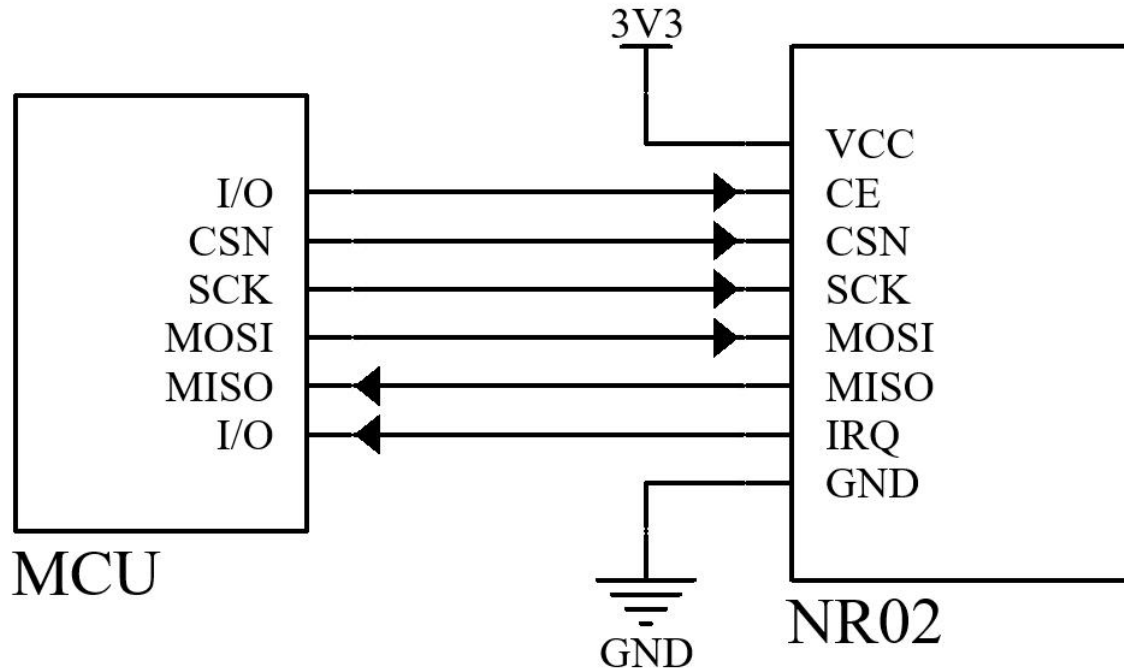


Figure 2: Typical application circuit

3. Electrical characteristics, RF characteristics, and reliability

3.1. Specifications

Table 5: Specification sheet

Main parameters	Description	Notes
Open space distance	300M	Clear open environment, air speed 250kbps, transmit power +7dbm
Communication interface	SPI	0-10Mbps
Crystal frequency	16MHz	-
FIFO	32Byte	-
Modulation	GFSK	-
Encapsulation method	Pin type	-
Interface mode	2.54mm 2*4P	-
Dimensions of form	15×25mm	-
Rf interface	PCB board antenna/external antenna	The equivalent impedance is about 50 ohms

3.2. Electrostatic protection

In the application of modules, due to the static electricity generated by human body static electricity and charged friction between microelectronics, it may cause some damage to the module through various ways, so ESD protection should be paid attention to. ESD protection measures should be taken in the process of research and development, production, assembly and testing, especially in product design. For example, at the interface of the circuit design and the point susceptible to electrostatic discharge damage or influence, anti-static protection should be increased, and anti-static gloves should be worn in production.



Table 6: Table of ESD tolerant voltage of module pins

Test interface	Contact discharge	Air discharge	Units
VBAT and GND	+4	+8	kV
Main antenna interface	+2.5	+4	kV

4. Mechanical dimensions and layout recommendations

This section describes the mechanical dimensions of the module, all dimensions are in millimeters;
All dimensions not marked with tolerances with tolerances of ± 0.3 mm

4.1. Modular mechanical ruler

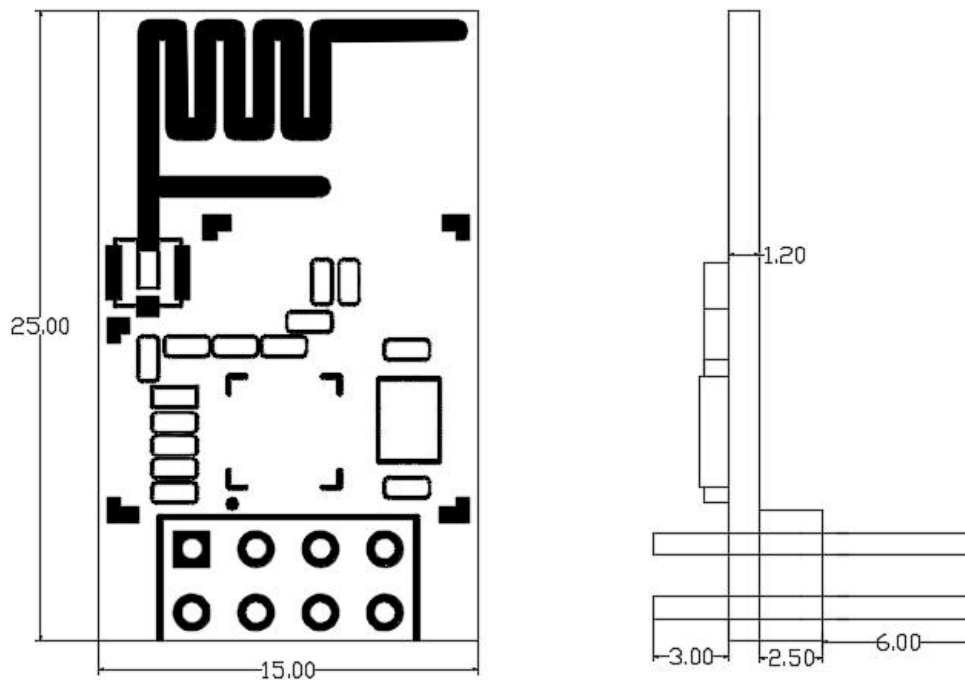


Figure 3: Top view and side view dimensions of the module



4.2. Recommended packaging

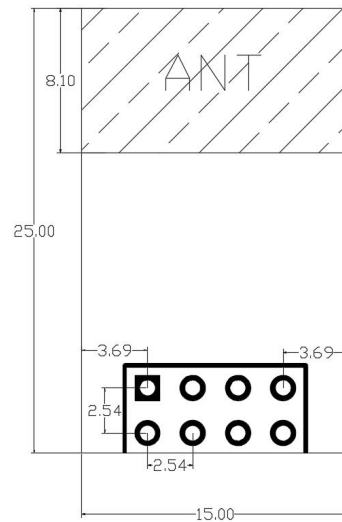


Figure 4: Proposed package size diagram

4.3. Top/bottom view of the module

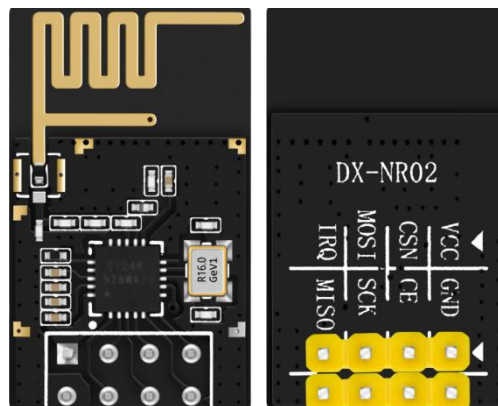


Figure 5: Module top view and bottom view

Notes

The above picture is for reference only, please refer to the actual product appearance and label information of the module.



5. Storage, production and packaging

5.1. Storage conditions

Modules are shipped in vacuum-sealed bags. The module has a humidity sensitivity class of 3 (MSL 3), and its storage is subject to the following conditions:

1. Recommended storage conditions: temperature $23\pm5^{\circ}\text{C}$ and relative humidity of 35~60%.
2. Under the recommended storage conditions, the module can be stored in a vacuum sealed bag for 12 months.
3. The unsealed workshop life of the module is 168 hours under workshop conditions of $23\pm5^{\circ}\text{C}$ and relative humidity below 60%. Under these conditions, the module can be directly used for reflux production or other high temperature operation. Otherwise, it is necessary to store the module in an environment with relative humidity less than 10% (for example, a moistureproof cabinet) to keep the module dry.
4. If the module is under the following conditions, it is necessary to pre-bake the module to prevent the PCB foaming, cracking and delamination after the module is wet and damp and then welded at high temperature:
 - Storage temperature and humidity do not meet the recommended storage conditions;
 - The module fails to complete production or storage according to Article 3 above after unpacking;
 - Vacuum packaging leakage, materials in bulk;
 - Before module repair;

5.2. Module baking treatment

- It needs to be baked at $120\pm5^{\circ}\text{C}$ for 8 hours at high temperature;
- The second baking module must be completed within 24 hours after baking welding, otherwise still need to be stored in the drying oven;



Notes

1. In order to prevent and reduce the occurrence of bad welding such as foaming and delimitation caused by moisture, the module should be strictly controlled. It is not recommended to be exposed to the air for a long time after opening the vacuum package.
2. Before baking, it is necessary to remove the module from the package and place the bare module on the high temperature resistant appliance to avoid high temperature damage to the plastic tray or reel; The second baking module must be completed within 24 hours after baking welding, otherwise it needs to be stored in the drying oven. Please pay attention to ESD protection when unpacking and placing the module, for example, wear anti-static gloves.

5.3. Reflow soldering

Use a printing scraper to print solder paste on the screen plate, so that the solder paste is leaked to the PCB through the opening of the screen plate, and the printing scraper strength needs to be adjusted appropriately. In order to ensure the quality of the module paste, the thickness of the steel mesh corresponding to the module pad part is recommended to be 0.1~0.15mm.

The recommended reflow soldering temperature is 235~250 °C, and the maximum temperature should not exceed 250 °C. To avoid damage to the module due to repeated heat exposure, it is highly recommended that customers do not attach the module until the first side of the PCB has been reflow soldered. The recommended furnace temperature curve (lead-free SMT reflow soldering) and related parameters are shown in the following chart:

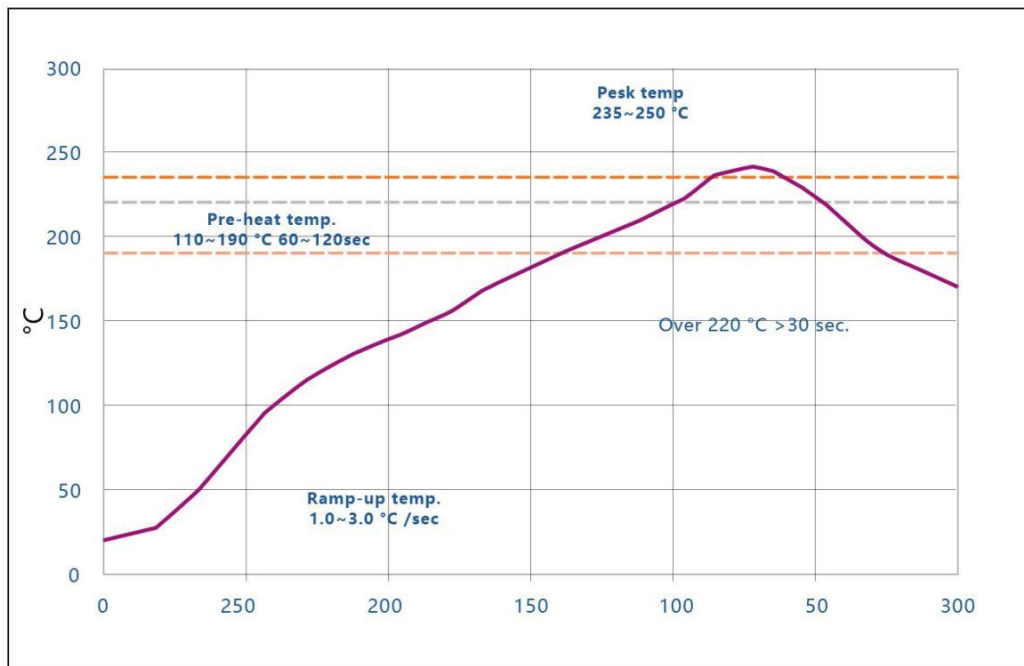


Figure 6: Recommended reflow soldering temperature curve

Table 7: Recommended reflow soldering temperature

Statistical name	Lower limit	Upper limit	Units
Slope 1 (target =2.0) is between 30.0 and 70.0	1	3	Degrees/SEC
Slope 2 (target =2.0) is between 70.0 and 150.0	1	3	Degrees/SEC
Slope 3 (target =-2.8) is between 220.0 and 150.0	-5	-0.5	Degrees/SEC
Constant temperature time 110-190 ° C	60	120	seconds
@220C reflux time	30	65	seconds
Peak temperature	235	250	Celsius
@235 ° C total time	10	30	seconds